Fine-tuning a Tsuboyama-pretrained ΔΔG model on FireProt

Experiment 08 · ddG_with_Boltz · concat features + antisymmetry · MLP · cross-dataset homology holdout.

> On a FireProt ≤500 test set of 25–27 homology-held-out proteins, **sequentially fine-tuning** a Tsuboyama-pretrained ΔΔG regressor on FireProt **does not reliably improve FireProt accuracy**: plain Tsuboyama-only transfer is the best in Pearson at the 30 % and 50 % thresholds, fine-tuning wins only at 90 %, and in rank correlation the two are within noise. The one robust effect is that training on FireProt **alone** forgets Tsuboyama. The winning recipe is therefore big-corpus pretraining + transfer, not fine-tuning on the small target set — consistent with the published literature and with this project's density-limited picture of the error.

1. Motivation

A Tsuboyama-trained model transfers to the independently curated FireProt dataset but is bounded by how densely the training data covers each ΔΔG value. A natural way to add coverage is to fine-tune on FireProt itself. The question is whether this helps FireProt, and whether it degrades the original Tsuboyama performance (catastrophic forgetting) — measured on both datasets under a split that controls for homology across them.

2. Methods

Splits. All wild-type sequences from both datasets are pooled and clustered by sequence identity (MMseqs2, 80% coverage) at 30/50/90%. Whole clusters are assigned to train or test, so no train/test pair — within or across datasets — exceeds the threshold; clusters spanning both datasets go to train. This yields disjoint *Tsuboyama-train / Tsuboyama-test / FireProt-finetune / FireProt-test* sets.

Model. 5-seed MLP ensemble on concat pair-track features with antisymmetry augmentation (the defaults established in experiment 07), evaluated in three conditions. **A (Tsuboyama-only):** trained on Tsuboyama-train. **B (FireProt-only):** a fresh model trained on FireProt-finetune alone, with its own input normalisation — the baseline for "how far does FireProt get on its own." **D (fine-tuned):** the pretrained model, warm-start continued on FireProt-finetune at a lower learning rate, reusing the Tsuboyama normalisation. All three are evaluated on Tsuboyama-test and FireProt-test.

3. Results

Identity	Pearson A	Pearson B	Pearson D	Spearman A / B / D
30%	**0.606**	0.510	0.598	0.579 / 0.514 / 0.594
50%	**0.599**	0.557	0.562	0.564 / 0.527 / 0.573
90%	0.514	0.521	**0.560**	0.533 / 0.515 / 0.605

Table 1. FireProt-test pooled correlation, A / B / D. Bold = best Pearson per row.

Identity	A (Tsu-only)	B (FP-only)	D (fine-tuned)
30%	0.801	0.721	0.795
50%	0.777	0.655	0.760
90%	0.607	0.721	0.503

Table 2. Tsuboyama-test pooled Pearson r, A / B / D (forgetting check).

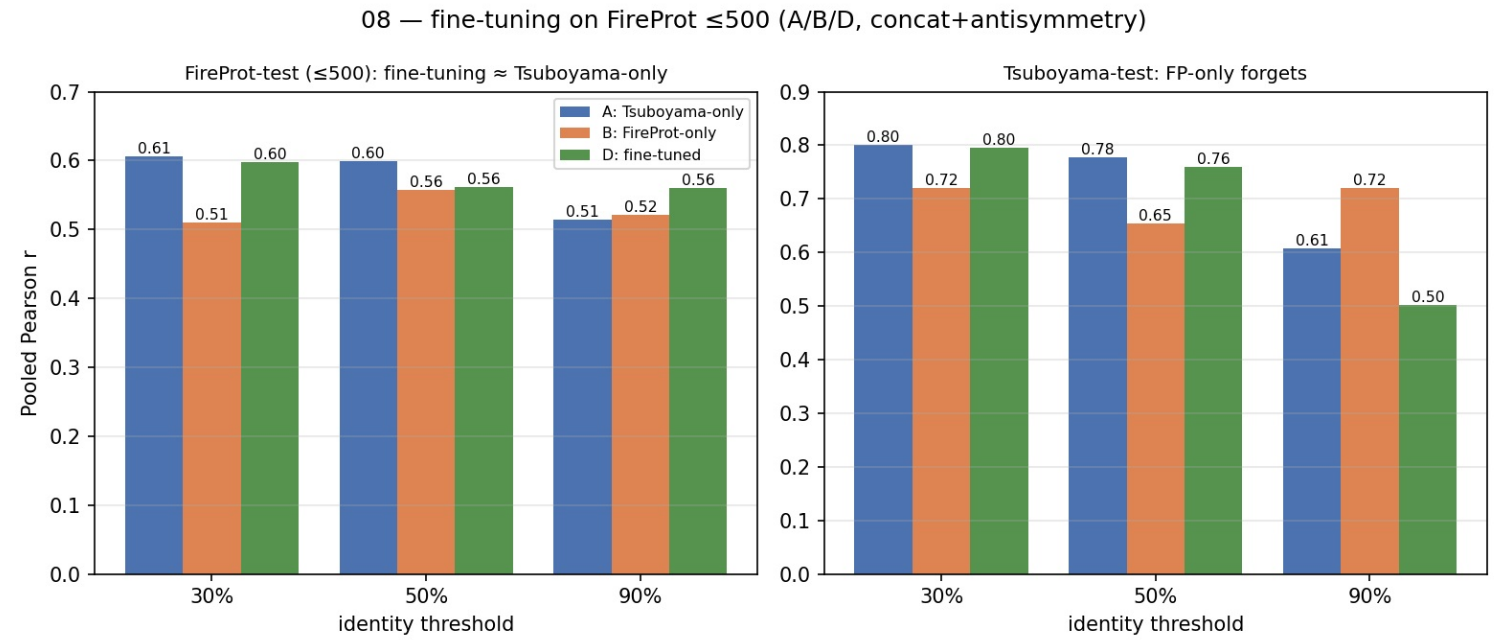

Figure 1. Pooled Pearson r for A / B / D. Left: FireProt-test — Tsuboyama-only (A) is best at 30/50 %, fine-tuning (D) only at 90 %. Right: Tsuboyama-test — FireProt-only (B) drops well below A/D (forgets); the 90 % A/D bars are a calibration outlier on that split.

4. Discussion

On this ≤500 test set — 25–27 homology-held-out proteins, ~2× the earlier ≤200 set — **fine-tuning does not reliably beat plain transfer**. In Pearson, Tsuboyama-only (A) is the best FireProt-test model at 30 % and 50 % identity; fine-tuning (D) wins only at 90 %, and in Spearman the two are within noise (D marginally ahead at all three). An earlier, smaller ≤200 experiment had suggested a modest fine-tuning gain; on the larger, less noisy test set that gain washes out, which is the more trustworthy reading. The result is consistent with the published literature (ThermoMPNN reports that fine-tuning on FireProt does not reliably help and that training on FireProt alone degrades). The one robust effect here is exactly that: the **FireProt-only baseline (B) forgets Tsuboyama** — its Tsuboyama-test correlation drops from ~0.80 to ~0.66–0.72 — because training on the small target set alone discards the broad Tsuboyama signal. The practical conclusion is that the predictor's accuracy is set by the pair-track features and the coverage of the large pretraining corpus, not by exposure to FireProt labels; big-corpus pretraining followed by transfer is the recipe, and fine-tuning on the small target set adds little.

5. Fine-tuning fixes calibration, not skill

One effect fine-tuning *does* have is on **calibration**, separate from correlation. A Tsuboyama-only model, trained on a narrow ΔΔG range, badly under-predicts magnitude on FireProt — its predictions span only ~43 % of the true spread (predicted-vs-experimental slope ≈ 0.26). Fine-tuning adds FireProt's wide-range examples, and the predictions decompress markedly: at the 50 % split the slope rises to ≈ 0.42 and the spread to ~76 % of the true range. Yet accuracy (RMSE) barely moves — because the optimal spread for a model with this correlation ($r \approx 0.56$) is only ~56 % of the true range, so fine-tuning actually *overshoots* the calibration optimum, trading systematic under-prediction for over-spread. The binding limit is the **ranking ceiling (r)**, set by the features, which fine-tuning does not move. In short: fine-tuning corrects the magnitude bias but not the predictive skill — the same conclusion from a different angle.